

fsb: Functional Specification

Version 1.0

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Abstract

fsb, the filesystem browser, is a local-only, read-only web view of a filesystem: one program that serves a browser interface on the loopback address and never changes what it shows. This document specifies its externally observable behavior: how it is started and configured, what it lists, searches and previews, which paths it refuses and how it refuses them, what the interface does with the keyboard, and what its HTTP interface answers. It does not say how any of this is built; that is the subject of the design specification. Every requirement has an identifier that other documents and the tests cite.

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1 Introduction

1.1 Purpose

This specification states what **fsb** does, precisely enough that a person could write an equivalent program from it and a tester could check the program against it. A statement belongs here if a user, or a tester who cannot see the source, could confirm it. How the behavior is achieved, and why that way was chosen, belongs to the design specification.

1.2 Product summary

fsb (“filesystem browser”) serves a web interface for browsing the user’s own files on macOS. It listens on the loopback address only, opens the interface in a browser, and lets the user list folders, filter and search by name, and preview files. It never writes to, renames, moves or deletes anything it serves. Two rule files let the user hide clutter (*ignore*) and permanently block sensitive locations (*deny*); a built-in set of deny rules for known credential locations is active even when no file exists.

1.3 Goals

- G1** Fast, comfortable browsing of an entire home folder in a browser, including folders with 100,000 or more entries.
- G2** *Local only*: no exposure beyond the loopback interface, with defenses against attacks that a web page can mount on a local server.
- G3** *Read-only*: nothing that **fsb** serves can be altered through it.
- G4** Explicit, testable exclusion semantics: a two-tier hide and deny model.
- G5** Simple distribution: a single binary.
- G6** Growth in stages (crawl, walk, run) without redesigning the security core.

1.4 Non-goals

- Remote access, sharing, multiple users, or any authentication beyond the per-launch credential of Section 7.
- Any operation that changes files: upload, rename, move, delete, edit.
- Platforms other than macOS in this version.
- Replacing Finder or a terminal as a file manager.
- Running as a background service.

1.5 Users

The primary user is the author, on their own Mac. Secondary users are developers and power users who install the program from a public repository and need to trust it after reading its documentation: the manual page, the README, and the security policy.

1.6 Terms

Root A folder that **fsb** serves. Nothing outside the roots is reachable.

Entry A file, folder or link inside a folder.

Ignore (hide) A rule that removes an entry from listings and search results. The entry stays reachable by its path.

Deny (block) A rule that makes a path unreachable by any means. A denied path is answered exactly as a path that does not exist.

Core rules The built-in deny rules for known credential and secret locations.

Launch URL The single-use address that **fsb** prints at startup.

Session The browser's authenticated state after the launch URL has been used once.

Cloud-only file A file whose contents are stored only in a cloud service and not on this machine (an “optimized storage” placeholder).

1.7 Conventions

- A requirement is written *shall*. Each has an identifier such as [CLI-4](#), which is a link in the PDF; the identifiers are stable and are cited by tests and by the other specifications. An identifier is never reused.
- “The user” is the person at the keyboard; “the browser” is the browser they opened the interface in.
- Paths are absolute unless stated. “Case” and “the same name” are defined in [Section 3.1](#).
- Limits are collected in [Section 9](#).

1.8 Related documents

Design specification How the behavior is achieved, and why.

Algebraic specification The laws that the access rules and the pure functions obey, each tied to an executable check.

Security policy The threat model, the guarantees, the known limitations, and how they were tested.

Manual page `fsb(1)`: the command line, files and exit status, in the form users expect.

2 Invocation and Configuration

2.1 Command line

CLI-1 The command is `fsb [flags] [path]`. With no flags and no path it serves the user’s home folder.

CLI-2 The optional `path` names the first root. More than one path argument is an error. A path that does not exist, or is not a folder, is an error. Symbolic links in a root are resolved, so a root is always a real location.

CLI-3 `-root PATH`, which may be repeated, adds further roots (for example `/Volumes/Data`). All roots are equal: each is listed as a place the user can switch to when there are several. A folder given more than once, under any spelling (a trailing slash, a symbolic link, `-root ~` when the home folder is already the default root), is served once.

CLI-4 The folder `/` is refused as a root unless `-allow-system-root` is given; the message says so. The decision is made on the folder itself, after symbolic links are resolved, not on how it was spelled: a path argument, a `-root` or a home folder that is a symbolic link to `/` is refused too. Rules ([Section 6](#)) apply to a root of `/` as to any other root.

CLI-5 By default the launch URL is opened in the user’s default browser. `-no-open` only prints it. `-browser NAME` opens it in the named macOS application instead (for example `"Google Chrome"`).

A name that begins with `-` or contains a NUL, carriage return or newline is refused before anything starts. If the browser cannot be opened, `fsb` says so, keeps running, and the printed URL remains usable. On systems other than macOS the URL is only printed.

CLI-6 `-debug` enables verbose logging on standard error and makes a request for a denied path answer with a body that names the matching rule (status still 404). No other flag or setting reveals which rule matched.

CLI-7 `-init` writes the default rule files (Section 2.5) and exits. It creates the folder `~/.config/fsb` with mode 0700 if needed, creates each file with mode 0600, and *never overwrites* a file that exists: it reports that the file was left unchanged. The rule files are the only files `fsb` creates that the user is meant to edit; its other files are listed in CFG-5 and CFG-6.

CLI-8 `-h` or `-help` prints the usage text and exits with status 0. An unknown flag prints the error and the usage text and exits with status 1.

2.2 Startup output

CLI-9 On success `fsb` prints, on standard output, one line naming the roots it serves and the loopback address and port, one line giving the launch URL, and one line saying how it is stopped (Control-C, or `-background`; under `-background` the line instead names its process and `fsb -stop`). The port is the one remembered from the last successful start (CFG-5), or one chosen by the operating system when nothing is remembered; `-port` (CLI-15) overrides it for one run.

CLI-10 The launch URL has the form `http://127.0.0.1:PORT/PREFIX/?token=TOKEN`, where PREFIX and TOKEN are random values generated at each start (Section 7). It works once; opening it a second time is refused.

CLI-11 If one or more core deny rules are not in effect (RUL-16), `fsb` prints a warning on standard error at startup naming the paths that are therefore browsable.

2.3 Running and stopping

CLI-12 `fsb` runs in the foreground (or, with `-background`, detached: CLI-17) until it receives SIGINT (Control-C) or SIGTERM (as `fsb -stop` sends: CLI-18). It then stops accepting requests, allows requests in progress up to three seconds to finish, and exits with status 0.

CLI-13 The exit status is 0 after `-help`, `-init`, `-show-deny`, `-show-ignore`, `-status` (whether or not `fsb` is running), a successful `-background` start or `-stop`, and a normal shutdown, and 1 after any error, including `-stop` when no `fsb` is running. An error is reported on standard error as one line beginning `fsb:`, and no server is started.

CLI-14 `fsb` refuses to start, with a message that names the file and the line, if the deny file or the ignore file cannot be parsed (Section 6.2). It never starts with weaker rules than the user wrote.

CLI-15 `-port N` uses exactly port N on 127.0.0.1 for this run (0 lets the operating system choose). A value outside 0–65535 is refused before anything starts. The port given is not remembered (CFG-5), so a one-off choice never replaces the usual port.

CLI-16 `-show-deny` and `-show-ignore` print the deny or ignore rules that would be in effect, one per line, and exit without serving. A first comment line names their source: the rule file, or the built-in rules when there is no file. For deny rules a last comment line names any core rule the file leaves out ([CLI-11](#)). Given together, both lists are printed, deny first. A rule file that cannot be parsed is an error, as in [CLI-14](#).

CLI-17 `-background` starts `fsb` detached from the terminal (in a new session, so closing the terminal or pressing Control-C there does not stop it), prints the startup output of [CLI-9](#), and returns to the shell once it is serving. If startup fails, the error is reported as in [CLI-13](#) and nothing is left running. If an `fsb` is already running (as found by [CLI-18](#)), the start is refused with a message naming it. Output after startup (such as `-debug` logging) goes to the log file of [CFG-6](#).

CLI-18 `-stop` stops the running `fsb`: the one started with `-background` (from its process-number file, [CFG-6](#)), or else an `fsb` started any other way that is listening on the port remembered in [CFG-5](#). It sends SIGTERM, waits up to five seconds for the process to exit, and reports it. Only a process that is `fsb` is ever signaled: it must belong to the same user and have been started from an executable file named `fsb` (the name a process gives itself does not count), and a process taken from the process-number file must also have the start time recorded there. Anything else is ignored, so a stale file whose number now belongs to another process is harmless.

CLI-19 `-status` reports whether an `fsb` is running (found as in [CLI-18](#)) and, if so, whether in the background or the foreground, its address, process number, start time and terminal; then exits.

CLI-20 If the remembered port ([CFG-5](#)) is held by another program, `fsb` does not start and does not silently choose a different port. The message names the port and, when it can be found, the program holding it with its process number, start time and terminal; if that program is another `fsb`, it suggests `fsb -stop`. Otherwise it suggests `-port`.

CLI-21 `-version` prints the version of `fsb` and the commit it was built from, on standard output, and exits with status 0 without reading any file or starting anything.

2.4 Environment

ENV-1 The home folder is taken from the operating system's notion of the user's home (the `HOME` environment variable), then resolved to its real location, so that a home folder reached through a symbolic link or an alias is treated as the real folder. It determines the default root, the location of the configuration folder, and the meaning of `~` in rules.

2.5 Configuration files

CFG-1 Configuration is two optional text files in `~/.config/fsb/`: `ignore` (hide rules) and `deny` (block rules). Both are global; there are no per-folder rule files.

CFG-2 The files are read once, at startup. Changes take effect at the next start.

CFG-3 If `ignore` does not exist, the built-in ignore rules apply (`.DS_Store`, `.git/`, `node_modules/`). If `deny` does not exist, the core deny rules of [Section 6.3](#) apply. If `deny` exists it is authoritative: whatever it does not contain is not enforced, and a core rule that is missing is reported as in [CLI-11](#) and [RUL-14](#).

CFG-4 User preferences (which columns are shown, the width of the preview pane, and the like) are not configuration files: they are kept in the browser (Section 5.3) and never on disk. Browsers keep them per address, including the port, which is why the port itself is remembered (CFG-5).

CFG-5 `~/.config/fsb/lastport` holds the port of the last successful start that did not use `-port`. It is written after such a start (best effort: a failure to write it never stops `fsb`) and read at the next start. A missing or malformed file means that nothing is remembered and never stops `fsb`, unlike a malformed rule file.

CFG-6 While an `fsb` started with `-background` runs, `~/.config/fsb/pid` holds its process number; it is removed when that `fsb` stops. `~/.config/fsb/background.log` receives its output after startup and is replaced at each `-background` start. Both are created with mode 0600.

3 Browsing

3.1 Names and case

NAM-1 Wherever this specification says that two names are “the same name”, it means what the file system means. On the macOS file system in its default, case-insensitive form, two names are the same name when they are equal after Unicode normalization and *full* case folding. Under full folding, `ss` and the letter sharp `s`, `s` and the long `s`, `k` and the Kelvin sign, and the ligatures `ff`, `fi`, `fl`, `ffi`, `ffl` and `st` are the same as their spelled-out forms. A name that the file system resolves to a file is that file, and every rule, root check and search treats it as such.

NAM-2 A name that contains characters that would disguise it, namely control characters, bidirectional overrides and isolates, line and paragraph separators, the byte order mark, and tag characters, is displayed with each such character replaced by a visible marker naming its code point (for example `<U+202E>`). Ordinary text, including text in right-to-left scripts, is displayed unchanged. Links and downloads use the real name.

3.2 Roots and reachability

ROOT-1 Only the roots (CLI-2, CLI-3) are reachable. A request for anything outside them, including through a symbolic link that leads outside, is answered as a path that does not exist. Whether a path is inside a root is decided by the identity of the folder, not by how the path is spelled. (On a case-sensitive volume, a root named `Public` does not admit a different folder named `public`.)

ROOT-2 When there is more than one root, the interface offers a way to switch between them. The box *Go to path* (`Alt+G`) accepts an absolute path, or one beginning `~/`, and resolves `.`, `...`, doubled and trailing slashes. A path outside the roots produces a message that says the folder is not served and lists the roots. `~name` (another user’s home) is refused.

3.3 Listing

LST-1 Opening a folder lists its entries with the columns *Name*, *Size*, *Modified*, *Kind* and *Permissions*, and an optional last column, *Attributes* (META-3). A folder has no size shown. Size is right-aligned.

LST-2 The listing is delivered incrementally, so the first rows appear before a very large folder has been read, and only the rows on screen are drawn. A folder of 100,000 entries remains responsive (Section 9).

LST-3 Entries are sorted as the tool `eza` sorts by default: names compare without regard to case, digits inside names compare by value (`file2` before `file10`), and a leading dot sorts before letters, so `.bashrc` comes first. Folders are *not* grouped by default. Clicking a column header sorts by that column, and clicking it again reverses the order. Ties fall back to the name and, if two names still compare equal (`file1` and `file01`, whose digits compare by value), to their raw characters, so the order is total and does not depend on the order in which the file system returned the entries.

LST-4 The option *Folders first*, off by default, lists all folders before all files under every sort column and in both directions. A folder, having no size, sorts as smaller than any file when sorting by size. A modification time that cannot be read sorts as the oldest.

LST-5 A symbolic link is shown with a link marker. Its size, date and kind are those of its target. A link whose target does not exist is shown as broken. A link whose target is denied, or lies outside the roots, is not shown at all.

LST-6 Sockets, devices and pipes are listed but never opened.

LST-7 Entries whose names begin with a dot are hidden until the option *Hidden files* is turned on. This option is independent of the ignore file (RUL-2) and never reveals an entry that a rule hides or denies.

LST-8 The box *Filter this folder* (key `/`) narrows the listing as the user types, to entries whose names contain the text. Matching ignores case and normalization as in NAM-1, unless the option *Match case* is on, in which case only normalization is ignored. The filter finds the same names that search finds (SRC-3).

LST-9 The interface shows the current path as breadcrumbs, each a link to that folder, and a button that copies the path (also `Alt+C`, which copies the selected entry's path, or the folder's when nothing is selected).

LST-10 The location in the browser's address bar identifies the folder (in the fragment, `#/Users/me/docs`) and optionally an entry to select (`?select=name`), so a folder can be bookmarked and the browser's Back button works while `fsb` is running.

LST-11 Columns can be resized by dragging their edge (double-click, or Enter on the edge, fits the column to the visible rows; Left and Right resize by 10 pixels, with Shift by 50), reordered by dragging a header onto another or with Alt and the arrow keys, and shown or hidden from the *Columns* menu (Name is always shown). Widths are between 60 and 900 pixels. *Reset columns* restores the defaults. The menu closes on an outside click and on Escape. On a narrow window the table scrolls sideways and does not hide columns.

LST-12 Denied entries never appear in a listing (RUL-1), whatever options are set. The interface shows a badge saying that it is read-only.

LST-13 Opening a folder that is itself stored only in the cloud (not downloaded) reports that plainly and reads nothing, rather than listing it as empty or answering with an error that does not say why ([SEC-13](#)).

3.4 Search

SRC-1 The box *Search subfolders by name* (**Alt+S**; Enter runs the search) finds entries under the current folder whose names contain the text, at any depth.

SRC-2 Results stream in as they are found, shallowest first, so the nearest matches appear immediately. The search stops at 1,000 results or after 500,000 entries have been examined, and the interface says when a limit stopped it.

SRC-3 Search compares names as in [NAM-1](#) (case, accents and normalization are ignored), unless *Match case* is on. A search for **strasse** finds a file named with the sharp s.

SRC-4 Search never enters or reports a denied, hidden or cloud-only folder, never reports a denied or hidden entry, and does not follow symbolic links to folders (so it always terminates). A cloud-only folder is skipped rather than failing the whole search. Anything a search reports is also shown when the folder that contains it is listed. A folder that was queued for searching and, by the time its turn came, had been replaced by a symbolic link (or lies behind one) is not entered, because its entries would be judged under a name that is not where they are; it counts as not searched ([SRC-6](#)).

SRC-5 Choosing a result opens its folder with that entry selected. While results are shown, the last breadcrumb, Escape, and a *Clear search* control all return to the plain listing.

SRC-6 A search that could not look at everything it was allowed to look at says so, and its results are not presented as complete. This covers a folder that is there and allowed but could not be opened (a permission error, an input/output error), a folder that failed partway through being read, and a folder replaced by a symbolic link as in [SRC-4](#). A denied folder, which looks exactly like a missing one, and a folder that vanished are not counted, and neither is a cloud-only folder, which is skipped on purpose ([SRC-4](#)): completeness is relative to the searchable, allowed content.

4 Preview and Metadata

4.1 The preview pane

PRV-1 Selecting an entry shows a preview of it in a pane beside the list, together with its details ([META-1](#)) and extended attributes ([META-2](#)). The pane is shown or hidden with the option *Preview pane* or the Space key, and its width, at least 240 pixels and at most the window's width less 320 pixels kept for the list, is changed by dragging its left edge or from the keyboard. The pane grows when the window does, and shrinks if the window narrows below what it needs. It starts open only on a window wide enough to hold it beside the list.

PRV-2 Which preview is shown is decided from the *contents* of the file, never from its name alone. The name is only a hint about which check to try first: a file named **.png** that is not an image is shown as ordinary text or as a binary file, and a file with another name whose bytes are a PDF is a

PDF. The one exception is the Quick Look picture ([PRV-12](#)), which is chosen by name because macOS draws it and decides itself what the document is.

PRV-3 Text is shown from the first 64 KB of the file and is always displayed as text, never interpreted as markup. When the file is longer, the pane says how much is shown. A file is text only if that part is valid UTF-8 without control characters; a binary file is reported as such, with its size, and its bytes are not sent to the browser. An empty file is reported as empty.

PRV-4 Source code is shown with syntax highlighting, chosen from the name or extension, for these languages: Bash, C, C++, CSS, diff, Dockerfile, Go, INI and TOML, Java, JavaScript, JSON, Kotlin, Makefile, Markdown, PHP, Python, Ruby, Rust, SQL, Swift, TypeScript, XML and HTML, and YAML.

PRV-5 A complete JSON file is shown pretty-printed; a JSON file that was cut off by [PRV-3](#) is shown as highlighted source. A CSV or TSV file is shown as a table of its first 200 rows, and the pane says when there are more.

PRV-6 A file whose leading bytes identify it as PNG, JPEG, GIF or WebP is shown as an image. Files of any other image type are not rendered: in particular SVG and HTML are never rendered, because they can carry script.

PRV-7 A Markdown file (name ending `.md` or `.markdown`) is shown rendered by default, with a *Source* toggle to see the text. Rendering supports headings, emphasis, lists, task lists, tables, code, block quotes and rules. Raw HTML in the file is shown as text and never interpreted. A link to another file opens that file inside `fsb`, in its folder with the file selected; a link to a web address opens in a new browser tab without sending a referrer; a link of any other kind (`javascript:`, `data:`, `file:` and so on) is shown as plain text. An image that lies inside the roots is shown; a remote image is replaced by a note that it was blocked, so previewing a file never contacts anything outside the machine. At most 20 images of at most 5 MB each are loaded for one file, four at a time; an image is cut off as soon as it passes the size limit, whatever its server said it would send, and a bigger one is not shown.

PRV-8 A file that begins with `%PDF-` is shown in the browser's built-in PDF viewer. A file named `.pdf` that does not begin so is shown as ordinary text or as a binary file. If the viewer takes the keyboard focus on its own (as Chrome's does, for the password field of an encrypted PDF), the focus returns to the list, so the arrow keys keep moving through the folder; the user can still click or tab into the viewer to use it.

PRV-9 A file recognized by its bytes as a zip, tar or gzip-compressed tar archive (including `.jar`) is shown as a table of its contents: each member's name and size, the format, and the number of entries. Nothing is extracted. At most 1,000 entries are shown; the listing reports when it stopped at a safety limit (Section 9). A zip's directory is read record by record, so a directory that claims more entries than it holds cannot make `fsb` allocate for them, and a zip64 directory is honored whenever it is present. A plain tar is listed by skipping each member's body, so the work does not grow with the size of its contents. Listing stops when the request is abandoned. Member names are displayed as text with control and Unicode formatting characters replaced, cut after 4,096 bytes, with an ellipsis added, and are never used as paths. A file that has an archive name but is not one is shown as ordinary text or as a binary file.

PRV-10 For a file stored only in the cloud, every preview reports that it is not downloaded and reads nothing ([SEC-13](#)).

PRV-11 A symbolic link is previewed as what it points to, and the pane names the target.

PRV-12 A Numbers, Pages, Word, Excel or PowerPoint document (`.numbers`, `.pages`, `.docx`, `.doc`, `.xlsx`, `.xls`, `.pptx`, `.ppt`), whether a single file or a package folder, is shown as the picture of its first page that macOS Quick Look draws, as Finder shows it, with a note saying so. It is subject to every rule of an ordinary preview. Quick Look draws a private copy that `fsb` reads, never the user's own file, so it cannot follow a Finder alias. A cloud-only document, or a package with a cloud-only part, is not copied; parts of a package that a listing leaves out (denied files, links, hard links to denied files) are left out of the copy. Copying is given 20 seconds and drawing 5; both are canceled when the request is abandoned. The 20 seconds is a cooperative limit: it is checked between reads and once more when a read returns, so work that ran out of time fails instead of succeeding, but it cannot interrupt a read that the file system is itself holding up. When there is no picture, the pane shows what it would show without one ([PRV-3](#), or *Folder* for a package). Keynote documents (`.key`) are denied by a core rule (Section [6.3](#)) and are not shown.

4.2 Hover peek

HOV-1 Holding the pointer over a readable file for about 400 ms shows a small bubble with its first 20 or so lines, as plain text. Binary files, folders, empty files and cloud-only files produce no bubble.

HOV-2 The bubble is dismissed by moving the pointer away, scrolling, or pressing a key. Results are remembered so that hovering again does not re-read the file, and the number of simultaneous requests is limited. The option *Hover previews* turns the feature off; the choice is remembered (Section [5.3](#)).

HOV-3 Hover peek passes through the same rules as every other access. Because hovering reveals contents more readily than choosing, files that commonly hold secrets are denied by default (Section [6.3](#)) and therefore never produce a bubble.

4.3 Details and extended attributes

META-1 The pane lists the entry's path, kind, size (in a readable unit and in bytes), modification time and permissions, the target of a symbolic link, and whether the file is stored only in the cloud.

META-2 Extended attributes are listed with their values. A value is shown as text if it is valid printable UTF-8 and as hexadecimal otherwise (labelled *hex*, the label before the digits). At most 64 attributes are shown, and a value larger than 4 KB is shown by name and size only. An attribute that another program changes while it is being read is read again, a few times; one that does not settle is shown by name only. Attributes are available on macOS and Linux; elsewhere none are shown.

META-3 The optional *Attributes* column, hidden by default and placed last, shows attribute *names* only, for example `com.apple.quarantine`. It is fetched only for the rows on screen and remembered, and is never part of the listing itself, so it cannot slow a large folder.

5 Keyboard and Preferences

5.1 Keys

KEY-1 The interface can be operated entirely from the keyboard. Navigation keys (arrows, Home, End, Page Up, Page Down, Space) are matched by the physical key, so they work whatever modifier is held; so are the Alt shortcuts below, because macOS changes the character that Option and a letter produce (Option+C types “©”). Other letters and punctuation are matched by the character they produce, so they follow the user’s keyboard layout. No shortcut acts while the user is typing in a text box, except Escape, which leaves it. The arrow keys that resize a column or the preview pane, when its edge has the focus, do only that: they do not also move the selection or change the folder.

KEY-2 The following keys are defined.

Key	Action
Up, Down, PgUp, PgDn, Home, End	Move the selection
Enter or Right	Open the selected folder, selecting its first entry (a search result opens in its folder, with it selected)
Left or Backspace	Up one folder, reselecting the one just left
Space	Show or hide the preview pane
/	Filter this folder
A letter or digit	Jump to the next entry whose name starts with what has been typed (KEY-3)
Alt+S	Search subfolders by name (Enter runs, Escape clears)
Alt+G	Go to a path
Alt+C	Copy the selected path (or the folder’s)
Escape	Clear the search, close a menu, leave a text box
Alt+Left, Alt+Right on a column header	Reorder the column

KEY-3 Type-ahead: typing a letter or digit selects the next entry, after the current one and wrapping around, whose name starts with it, compared as the filter compares names ([LST-8](#)). Letters typed within 800 ms of each other extend the prefix; repeating the same letter cycles through the entries that start with it. No plain letter is reserved for a shortcut, so every name can be reached this way.

5.2 Mouse

MOUSE-1 A plain left click on a row selects it, exactly as moving to it with the keyboard would; a click on a folder also opens it. A plain click on a file never downloads it. A modified click (with Command, Control, Shift or Alt) or the browser’s context menu on a file name still offers the browser’s own download, which is the explicit way to download ([API-4](#)).

5.3 Preferences

PREF-1 The following are the user’s preferences and are kept in the browser’s local storage, per browser and never on the server or on disk elsewhere: whether the preview pane is open and how

wide it is, whether hover peek is on, “Folders first”, “Match case”, and the layout of the columns (order, widths, which are hidden).

PREF-2 Preferences are optional. If local storage is missing, blocked, corrupt or from an older version, each value falls back to its default on its own and the interface works normally.

6 Exclusion Rules

6.1 Two tiers

	<i>ignore</i> (hide)	<i>deny</i> (block)
Purpose	Reduce clutter	Never serve
File	<code>~/.config/fsb/ignore</code>	<code>~/.config/fsb/deny</code>
Listings and search	Entry omitted	Entry omitted
Direct access (read, download, preview, details)	Allowed	Blocked, answered as 404
Negation (!)	Allowed	Refused
Defaults	<code>.DS_Store</code> , <code>node_modules/</code>	<code>.git/</code> , The core rules (Section 6.3)

RUL-1 A path matched by a deny rule is not listed, not found by search, not readable, downloadable, previewable or describable, and not reachable through any other endpoint. A request for it is answered exactly as a request for a path that does not exist: the same status and the same body, except under `-debug` (CLI-6). Deny always wins over ignore; no ignore rule, including a negation, can expose a denied path.

RUL-2 A path matched by an ignore rule is omitted from listings and search results but remains reachable by its path. Entering an ignored folder on purpose (by typing its path or following a link) lists its contents; only the folder’s own entry, in its parent’s listing, is hidden.

RUL-3 A denied folder denies everything inside it. Once a folder is excluded, no later rule can bring anything inside it back.

RUL-4 Symbolic links and aliases do not change what a rule means. A path is refused if *either* the path as requested *or* the real location it resolves to is denied; a link whose target is denied is refused, and is not listed. The children of a folder are judged by the folder’s real location, however it was reached. On macOS the aliases of the data volume are the same as the ordinary paths: `/System/Volumes/Data/Users/me` is `/Users/me`, in any letter case and any spelling that the file system equates (NAM-1).

RUL-5 Every spelling that the file system resolves to a denied file is refused. This includes other letter cases, decomposed accents, characters that fold to ASCII (NAM-1), doubled or trailing separators, dot segments, symbolic-link and firmlink aliases, and resource-fork names such as `file/..namedfork/rsrc`.

RUL-6 A file with more than one name (a hard link) is judged by its identity: if any of its names is denied, it is refused under every name and is not listed. The names are found under the roots and under the credential locations of every home folder on the machine, using an index built in

the background at startup (about 20 seconds for a home folder of 700,000 files). A file with several names is served only when all of the following hold: the index has found as many distinct names as the file has links (names are compared as the file system compares them, [NAM-1](#), by folder and name, so one name reached two ways counts once); none of them is denied; and, when the file is used, each recorded name is still a real path, with no symbolic link along it, to this very file. Otherwise it is refused rather than guessed: while the index is not ready, when a name lies outside every location or in a folder that could not be read, when the walk stopped at its limits ([Section 9](#)), and after the names changed (a link added or removed, a file or folder renamed or moved). The index is walked again, at most as often as its cooldown allows, and a file found short of names is owed one more walk, which is spent only when that walk starts; if it still finds fewer names than links, the file stays refused until something changes. A listing or a search reuses one verdict about a file for at most one second (so a folder of many names of one file costs one check, not one per name); after that the recorded names are checked again. This is a memo reuse interval, not a guarantee about the age of everything the user sees: entries already sent are not retracted, and a request to open the file never uses the memo. A denied name that lies outside every root and every credential location is not detected.

RUL-7 A permission error on a path that lies inside a denied place is reported as not found, so that an unreadable file in a denied folder cannot be told from a missing one. A permission error elsewhere is reported as such (status 403) and stays visible to the user.

RUL-8 Denying is decided before anything is read. A denied path never causes a cloud-only file to be downloaded.

6.2 Rule syntax

The syntax is that of `.gitignore`, with the additions and restrictions below.

RUL-9 A line is blank, a comment (first character `#`), or a rule. Surrounding white space is ignored. A leading byte order mark and any line ending (LF, CR LF or bare CR) are accepted. In a rule, `*` matches any characters except `/`, `?` one character except `/`, `[abc]` and `[!abc]` a class (which never matches `/`), and `**` any number of folders. A backslash makes the next character literal (`\#` for a literal `#`). A trailing `/` makes the rule apply to folders only.

RUL-10 A pattern with no `/` matches a name at any depth. A pattern that begins with `~/` is relative to the home folder; one that begins with `/` is absolute; one that contains a `/` elsewhere is relative to the home folder; and one that begins with `**/` matches at any depth.

RUL-11 Matching ignores letter case and normalization exactly as in [NAM-1](#), and ignores trailing white space in a name (macOS permits names such as `id.pem` and `.env` followed by a tab), and `**` matches names that contain newlines.

RUL-12 A rule file must work as written or fail loudly. `fsb` refuses to start, naming the file, the line and the problem, for: negation in the deny file; an empty pattern; a trailing backslash; a `/` inside `[...]`; `~name` (another user's home; only `~` and `~/...` are supported); a pattern wrapped in quotes; an inline comment (a `#` after white space; write `\#` for a literal one); an invisible or formatting character; and a character that looks like a slash but is not one.

6.3 Core and optional deny rules

RUL-13 The core deny rules cover locations and files that hold credentials or secrets. They are active by default and built in, so they apply even when no deny file exists. `fsb -init` writes them, uncommented, into the deny file so that the user can see them. They match at any depth, so the same folders in a backup or clone of the home folder, in another user's home, or under a wrong HOME are protected too.

RUL-14 The core rules are:

Folders	Files
<code>.ssh/</code> , <code>.aws/</code> , <code>.gnupg/</code> , <code>.kube/</code>	<code>.netrc</code> , <code>.git-credentials</code> , <code>.pgpass</code>
<code>**/.config/gh/</code>	<code>.env</code> , <code>.env.*</code>
<code>**/Library/Keychains/</code>	<code>*.pem</code> , <code>*.key</code>
<code>**/Library/Application Support/Google/Chrome/</code>	<code>id_rsa</code> , <code>id_dsa</code> , <code>id_ecdsa</code> , <code>id_ed25519</code> ,
<code>**/Library/Application Support/Firefox/</code>	<code>id_ecdsa_sk</code> , <code>id_ed25519_sk</code>
<code>**/Library/Safari/</code> , <code>**/Library/Cookies/</code>	<code>*.p12</code> , <code>*.pfx</code> , <code>*.ppk</code> , <code>*.jks</code> , <code>*.keystore</code>
	<code>*.kdbx</code> , <code>*.keychain</code> , <code>*.keychain-db</code>

Public key files (`id_rsa.pub` and the like) are not denied. The list is versioned with the release, and additions are called out in the release notes.

RUL-15 The secret-file patterns are deliberately broad and have known costs. `*.key` also matches Keynote presentations; `.env` also matches a folder named `.env` (such as a Python virtual environment) and everything in it; and `.env.*` also matches committed templates such as `.env.example`. A user who needs any of them removes the pattern from the deny file and accepts the warning of [RUL-16](#).

RUL-16 When a deny file exists but does not contain one or more core rules (they were removed or commented out, or a release added a rule the file predates), `fsb` (a) prints the warning of [CLI-11](#), for example `fsb: warning: core deny rule(s) disabled: .ssh/, .aws/; these paths are browsable.`, and (b) shows a persistent banner in the interface with the same message. Both disappear once every core rule is in effect. Rules are compared as names are ([NAM-1](#)), so `.SSH/` counts as the core rule `.ssh/`, but a rule that differs in meaning does not. The condition cannot be suppressed by any flag. `-show-deny` ([CLI-16](#)) also names the missing rules.

RUL-17 The optional rules, shipped commented out for the user to enable, are situational or prone to false positives: `~/ .docker/config.json`, `~/ .npmrc`, `~/Library/Mail/` and `~/Library/Messages/`.

7 Security Behavior

This section states what `fsb` does to protect the user, as behavior that can be observed. The reasoning, and the ways it was tested, are in the design specification and the security policy.

7.1 Reach

SEC-1 `fsb` listens only on the loopback address `127.0.0.1`. No flag, setting or environment variable changes this, and it is not reachable from another machine.

SEC-2 A request is refused (status 403) unless its `Host` header is exactly `127.0.0.1:PORT` or `localhost:PORT`, so that a web page on another name that resolves to the loopback address (DNS rebinding) cannot use it.

SEC-3 A request is refused (status 403) if it carries an `Origin` or `Referer` from any other origin, if its `Sec-Fetch-Site` says it was started by another origin or another site (including a page on another port of the same host), or if its `Origin` is the opaque value `null`. No cross-origin access is permitted.

SEC-4 Only the methods `GET` and `HEAD` are answered; anything else gets status 405. `fsb` never modifies, creates, renames, moves or deletes anything that it serves. The only files it writes are its own, in `~/.config/fsb/`: the rule files of `-init` ([CLI-7](#)), which never overwrites one, the remembered port ([CFG-5](#)), and the process-number and log files of a background instance ([CFG-6](#)). Before writing one of them `fsb` checks that it is an ordinary file with a single name that belongs to the user, and opens it without blocking and without emptying it first, so that a link, a pipe or another user's file placed there cannot make `fsb` overwrite something else.

7.2 Credential

SEC-5 Access requires the session, which is established as follows. At each start `fsb` generates a random launch token (at least 128 bits), a random session value, and a random path prefix (at least 128 bits). The launch URL carries the token. The first request that presents the correct token is answered with a redirect that removes the token from the address bar, and sets the session cookie. The token then no longer works, so a copy of the URL left in shell or browser history is useless.

SEC-6 The session cookie is `HttpOnly` (scripts cannot read it) and `SameSite=Strict`. Its path is the random prefix.

SEC-7 Browsers scope cookies by host and never by port, so a plain cookie would be sent to every other web server on `127.0.0.1` that the same browser visits. For that reason every URL of `fsb` lies under the random prefix (`/PREFIX/`), and a request whose path is not under it is refused (status 403) like an unauthenticated request, whatever cookie it carries. The interface forms every request relative to its own location, so the cookie accompanies exactly the requests that need it.

SEC-8 A request without a valid session is refused with status 403 and the same body, whatever it asked for. The launch token is never written to the log.

7.3 Responses

SEC-9 Every response carries `X-Content-Type-Options: nosniff`, `Cache-Control: no-store`, `Referrer-Policy: no-referrer`, `Cross-Origin-Resource-Policy: same-origin`, and, for the interface itself, `X-Frame-Options: DENY` and a content security policy that permits

only resources of its own origin, forbids framing, and forbids form submission and base URLs. The interface cannot be framed by another page.

SEC-10 A file is never displayed from the interface’s own origin except as one of the previews of Section 4. Downloading (API-4) always answers with `Content-Type: application/octet-stream` and `Content-Disposition: attachment`, so a browser saves the file and never renders it.

SEC-11 Inline responses other than the interface itself are limited to: raster images whose type is confirmed by their bytes (PNG, JPEG, GIF, WebP), which are sent with a policy that sandboxes them and allows nothing else; PDF documents whose bytes begin with `%PDF-`, which may be framed only by the interface’s own pages; PNG pictures that macOS Quick Look draws of iWork and Office documents (PRV-12), sent like the raster images; and the fixed page that renders Markdown, which runs in a sandboxed frame that has no access to the session or the interface, may run only its own fixed script, and may load nothing but images given to it as data. File names are never trusted.

SEC-12 fsb makes no outbound network connection, sends no telemetry, and does not update itself. Previewing a file never loads anything remote (PRV-7).

SEC-15 fsb runs no other program on the user’s files except Apple’s `qlmanage`, to draw a Quick Look picture (PRV-12). It is never given the user’s own file: it is given a private copy of a document that every other endpoint would serve, which fsb read after the same checks, as a single argument and never through a shell. The copy carries no extended attributes or resource fork. It is stopped after 5 seconds, at most two run at once, and one whose request is abandoned is stopped. Copying the document for it is limited to 20 seconds, cooperatively (PRV-12), and also stops when the request is abandoned.

SEC-13 fsb never causes a cloud-only file or folder to be downloaded. The details, the hover peek, every preview, and the archive listing check whether a *file* is stored only in the cloud, and listing and search check whether a *folder* is (a folder can be a cloud-only placeholder in its own right, not only the files inside it); each reports “not downloaded” (status 409) instead of reading it. The one exception is an explicit download by the user of a cloud-only file (API-4), which is the user’s request for it; there is no equivalent for a folder, since fsb never downloads or extracts one. The same holds for the hard-link index walk (RUL-6) and the copy of a package made for Quick Look (PRV-12): neither enters a cloud-only folder, and a package with a cloud-only part is refused.

SEC-14 A path that is refused for any reason other than a permission error on a non-denied path is answered as not found (status 404) with one and the same body. In particular a denied path, a path outside the roots, a path that does not exist, and a path that is not a regular file or folder cannot be told apart. Timing is not equalized: a denied path can be answered slightly faster than a missing one.

8 HTTP Interface

The browser interface talks to fsb over HTTP. This interface is described so that it can be tested and reviewed. It is *internal*: it may change between versions, it is not a supported programming interface, and it is refused to anything that does not hold the session (Section 7).

All paths in this section are relative to the random prefix (SEC-7): the interface is at /PREFIX/, and the endpoints at /PREFIX/api/.... Every endpoint takes the request path in the query parameter `path`, an absolute path, decoded once; a literal `%2e%2e` is just part of a name.

8.1 Endpoints

API-1 The interface answers the endpoints of the following table, and no others. All are `GET` (or `HEAD`).

Endpoint	Parameters	Answer
/		The interface (static files).
api/status		JSON: <code>readOnly</code> (always true), <code>coreDenyMissing</code> (list), <code>roots</code> , <code>home</code> .
api/list	<code>path</code>	The entries of a folder, streamed (API-2).
api/file	<code>path</code>	The file's bytes as a download (API-4).
api/head	<code>path</code> , <code>bytes</code>	A classified look at the start of a file (API-5).
api/meta	<code>path</code> , <code>values</code>	Details and extended attributes (API-6).
api/preview	<code>path</code>	A raster image for inline display.
api/pdf	<code>path</code>	A PDF for inline display.
api/archive	<code>path</code>	The table of contents of an archive.
api/quicklook	<code>path</code>	The Quick Look picture of an iWork or Office document (API-9).
api/search	<code>path</code> , <code>q</code> , <code>case</code>	Name matches under a folder, streamed (API-3).
api/mdframe		The fixed page that renders Markdown.

API-2 `api/list` answers with newline-delimited JSON: a line `{"path":P}`, then lines `{"entries":[...]}` in the order the file system gave them (the client sorts), then `{"done":true}`, or, if the listing broke after it had started, `{"error":"listing interrupted"}` instead. A client treats a stream that ends with neither as broken, never as a complete listing. An entry has `name`, `isDir`, `size`, `modTime` and `mode`, and, when true, `isSymlink` and `broken`. A denied, hidden or unreachable entry is not present. A path that is not a folder, is denied, or does not exist is answered 404.

API-3 `api/search` answers with newline-delimited JSON: a line `{"path":P,"query":Q}`, then lines `{"matches":[...]}` as results are found (each with `path`, `rel`, `name`, `isDir`, `size` and `modTime`), then `{"done":true,"visited":N,"truncated":B,"incomplete":B}`, or `{"error":"search interrupted"}` instead. `incomplete` is true when part of what the search was allowed to look at could not be searched (SRC-6); the interface then says so. A client treats a stream that ends with neither line as broken. `case=1` makes matching case-sensitive. The limits are those of SRC-2. An empty query finds nothing.

API-4 `api/file` answers a regular file with `Content-Type: application/octet-stream` and `Content-Disposition: attachment; filename=...` and supports range requests (status 206) for large files. A folder is answered 404.

API-5 `api/head` reads at most `bytes` bytes (default 2048, at most 65,536; a value outside 1 to 65,536 is answered 400) and answers JSON with `kind` (`text`, `binary`, `empty` or `dataless`), `size`, and, for text, `text` and `truncated`. Text is returned only if it is valid UTF-8; a binary or cloud-only file's bytes are never returned.

API-6 `api/meta` answers JSON with `name`, `path`, `isDir`, `size`, `modTime`, `mode`, `isSymlink`, `symlinkTarget`, `dataless` and `xattrs` (each with `name`, and, unless `values=0`, `value`, `encoding` (utf8 or hex), `size` and `large`).

API-7 `api/preview`, `api/pdf` and `api/archive` answer only for a file whose bytes are of the corresponding kind ([PRV-2](#)); any other file is answered 415. `api/preview` and `api/pdf` support range requests.

API-9 `api/quicklook` answers a PNG picture, sent like `api/preview` and never cached, for a document of one of the types of [PRV-12](#), file or package. It is answered 415 for any other type, for a document larger than 256 MB, and when Quick Look cannot draw the document in time; 409 for a cloud-only document or package part; and 404 for a denied or missing document. A package answers the same whether or not it holds parts that a listing leaves out. It does not support range requests.

8.2 Status codes

API-8 The status codes and their meanings are the following.

Status	Meaning
200, 206	Success (206: a range of the file).
303	The launch token was accepted; the session cookie is set (SEC-5).
400	A parameter is out of range (<code>bytes</code>).
403	No valid session, a wrong <code>Host</code> , a cross-origin request, or a URL outside the prefix (Section 7); or a permission error on a non-denied path (RUL-7).
404	Not found: the path does not exist, is denied, lies outside the roots, or is not a regular file or folder (SEC-14).
405	A method other than <code>GET</code> or <code>HEAD</code> .
409	The file is stored only in the cloud and was not read (SEC-13).
415	The file is not of the kind the endpoint serves (API-7).
416	A range that cannot be satisfied.
500	An internal error (the details are logged, not sent).

9 Limits and Non-Functional Requirements

9.1 Limits

LIM-1 The following limits apply. Each bounds the work or the amount of data that one request or one preview can cause, and each is reported to the user where it stops something.

Limit	Value
Search results / entries examined by one search	1,000 / 500,000
Text read for a preview	64 KB (PRV-3)
Bytes one <code>api/head</code> call may request	1 to 65,536 (default 2,048)
Rows of a CSV or TSV table	200
Lines in a hover bubble	about 20
Extended attributes shown / largest value shown	64 / 4 KB
Markdown images per file / size of each / fetched at once	20 / 5 MB / 4
Archive entries shown	1,000
Tar entries scanned / decompressed bytes read (tar.gz)	200,000 / 512 MiB
Zip: entries read / bytes of its directory, at which reading stops (checked between records, so the last may cross it)	200,000 / 64 MiB
Archive member name: bytes kept, then an ellipsis (3 bytes)	4,096
Column width	60 to 900 pixels
Preview pane width	240 pixels to the window's width less 320
Quick Look: time to copy / time per picture / pictures drawn at once	20 seconds / 5 seconds / 2 (PRV-12)
Hard-link verdict reused within one listing or search	1 second (RUL-6)
Quick Look: picture size / its bytes	1,200 pixels / 20 MB
Quick Look: document copied / entries in a package	256 MB / 5,000
Hard-link index walk (entries / time)	20,000,000 / 10 minutes
Time to finish requests at shutdown	3 seconds
Time <code>-stop</code> waits for <code>fsb</code> to exit	5 seconds (CLI-18)

LIM-2 When a limit stops a search, an archive listing or a text preview, the interface says so ([SRC-2](#), [SRC-6](#), [PRV-3](#), [PRV-9](#)); a limit never makes a listing silently incomplete.

9.2 Non-functional requirements

NFR-1 Listing a folder of 100,000 entries shall paint the first screen in under 500 ms on a recent Mac. *Measured* (2026-09-20, 100,000 files): first rows in about 450 ms, the full listing in 1.5 s, 31 rows drawn.

NFR-2 Idle memory shall stay under 50 MB, with no persistent index. *Measured*: 12.6 MB resident after start.

NFR-3 From launch to the URL being printed shall take under one second. *Measured*: 0.36 s.

NFR-4 The program shall be a single binary of under 20 MB with the interface embedded. *Measured*: 11.4 MB (2026-10-02).

NFR-5 Dependencies shall be few and reviewed, the module checksums pinned, and known vulnerabilities scanned for on every change.

NFR-6 The interface shall be operable from the keyboard (Section 4 and the key table), use semantic markup, and respect the system's light or dark appearance.

NFR-7 Rule matching shall cost no more than about 10 μ s per entry with the core rules, so that listing a folder of 100,000 entries adds about one second at most. *Measured: 9 μ s.*

10 Scope of This Version and Distribution

10.1 Stages

fsb is delivered in three stages that grow the interface without changing the security core: *crawl* (browse), *walk* (preview and search) and *run* (more). This version implements the crawl and walk stages, as specified in Sections 2 to 8. It has not been run against real cloud-placeholder files or in Firefox; those checks are still to be done.

10.2 Not in this version

The following are wanted but are not part of this version. They are recorded here so that they are not designed away.

FUT-1 Full-text and fuzzy search by delegating to the system’s Spotlight index, with every result checked against the deny and ignore rules before it is shown.

FUT-2 Live updates when files change.

FUT-3 Several panes, tabs, and saved views.

FUT-4 Version-control status overlays.

FUT-5 “Reveal in Finder” and “Open in editor”, restricted to a fixed list of actions, never building a command from a path. Excluded from this version because they would make the server start local processes at the user’s request; “Copy path” covers the need meanwhile. The one process this version starts is Quick Look’s `qlmanage`, a fixed program run on a private copy (SEC-15).

FUT-6 Derived data (an index, thumbnails, caches), if ever added, shall honor the deny rules when it is built and when it is queried.

10.3 Distribution

DST-1 The program shall be published in a public repository under the MIT license.

DST-2 Releases are source releases: a version tag in the public repository, installed with `go install github.com/rprimer/fsb/cmd/fsb@latest` or built with `make install` from a clone. No prebuilt binaries are provided, so none needs signing or notarization; a program built locally is not quarantined by macOS.

DST-5 The source shall provide `make install`, which installs the program and its manual page; the installation locations (`PREFIX`, `BINDIR`, `MANDIR`) shall be variables that the user can override, and `DESTDIR` shall stage the installation for packaging. The manual page goes in the `man1` folder of `MANDIR`.

DST-3 No Homebrew formula is provided. (Planned earlier; withdrawn, because maintaining one is ongoing work for a project that is finished.)

DST-4 The documentation shall explain the security model, the deny rules and their costs, and that macOS may ask for permission to read Documents, Desktop, Downloads and external volumes the first time.

Version 1.0 is released as source ([DST-2](#)), and the project is provided as-is: it is finished and not actively maintained.

A Where the Earlier Requirements Went

The product requirements document that preceded this specification numbered its requirements FR, ER, SR and NFR. It has been retired; this table records where each of its requirements now stands. Goals G1 to G6 and the non-goals are unchanged (Section 1). The earlier ER requirements that describe the construction of the access check (canonicalization, opening before verifying, package boundaries) are now in the design specification.

Earlier	Now
FR-1	CLI-5, CLI-9, CLI-10
FR-2, FR-2a, FR-2b	LST-1, LST-3, LST-8, LST-11, PREF-1
FR-3	LST-9, LST-10
FR-4	LST-2
FR-5	SEC-10, API-4
FR-6, FR-7	LST-7, RUL-1, RUL-2
FR-8	CLI-6
FR-9	LST-5, LST-6, RUL-7
FR-9a	CLI-11, RUL-16
FR-10	PRV-1 to PRV-11
FR-10a	HOV-1 to HOV-3
FR-11	KEY-1, KEY-2
FR-12, FR-12a, FR-12b	SRC-1 to SRC-5, LST-9
FR-13, FR-13a	META-1 to META-3
FR-14 to FR-18	FUT-1 to FUT-5
ER-3, ER-9, ER-10	RUL-4
ER-4, ER-12	NAM-1, RUL-11
ER-6	RUL-1, SEC-14
ER-7	FUT-6
ER-8	CFG-2, CFG-3, CLI-14, RUL-16
ER-11	RUL-7
SR-1 to SR-3	SEC-1 to SEC-3
SR-4	SEC-5 to SEC-8
SR-5	SEC-4
SR-6, SR-10	SEC-9 to SEC-11
SR-7, SR-8, SR-9	SEC-12, SEC-8, SEC-13
NFR-1 to NFR-6	NFR-1 to NFR-6
Section 11 (distribution)	DST-1 to DST-4